

1. Machine Translation – 7-mark Answer Machine Translation (MT) refers to the automatic conversion of text or speech from one natural language to another, and it is one of the earliest and most impactful applications of NLP. Modern MT systems have evolved from rule-based and statistical models to neural architectures, especially encoder–decoder frameworks supported by attention mechanisms and Transformer models. These systems learn semantic, syntactic, and contextual relationships from large multilingual corpora, allowing them to translate not only word-to-word but meaning-to-meaning by capturing long-distance dependencies and idiomatic expressions. The translation process generally involves tokenization, context representation in a latent vector space, alignment of source and target segments using attention, and decoding into fluent target sentences. MT is used in multilingual communication, cross-border business, real-time translation services, content localization, tourism, education, and international information sharing. Despite improvements, MT still faces challenges in low-resource languages, cultural references, ambiguity, and domain-specific terminology. Nonetheless, continuous advancements in neural models have made MT highly reliable and central to global digital communication.

2. Speech Recognition – 7-mark Answer Speech recognition, also called Automatic Speech Recognition (ASR), is the technology that converts spoken language into written text. It operates by analysing an audio signal, extracting acoustic features such as spectrograms or MFCCs, and mapping these features to phonetic or linguistic units using machine learning models. Traditional systems used Hidden Markov Models (HMMs) combined with Gaussian Mixture Models, but modern ASR relies heavily on deep neural networks, particularly recurrent networks, CTC-based models, and Transformer architectures. These models effectively capture temporal variations, coarticulation effects, speaker differences, and background noise, enabling more accurate transcription. A language model is usually integrated to ensure grammatically and contextually appropriate output. ASR supports numerous real-world applications, including virtual assistants, dictation tools, meeting transcription, voice-controlled devices, call-center automation, assistive technologies for differently abled users, and hands-free operations in vehicles and industrial systems. Challenges still exist in recognizing accented speech, code-mixed languages, noisy environments, and conversational spontaneity, but advancements in self-supervised learning and large speech datasets continue to improve performance.

3. Image Captioning – 7-mark Answer Image captioning is an interdisciplinary task that combines computer vision and natural language processing to generate descriptive sentences for images. The process begins with a vision model—traditionally CNNs and more recently Vision Transformers—that extracts high-level features representing objects, scenes, and spatial relationships within the image. These features are then provided to a language model, often implemented using RNNs, LSTMs, or Transformer decoders, which translates visual information into natural-language sentences. Attention mechanisms play a crucial role by allowing the model to focus on specific regions of the image while generating each word, improving accuracy and relevance. Image captioning has significant applications in accessibility services for visually impaired users, automatic generation of alt-text for websites, organization and retrieval of multimedia content, surveillance analysis, and digital content platforms. Despite improvements, the task remains challenging due to ambiguity in visual scenes, variation in descriptive styles, and the need for deep semantic understanding. Continuous integration of large-scale visual–text datasets and multimodal learning techniques has made image captioning one of the key advances in AI-driven multimedia understanding.

4. Spam Detection – 7-mark Answer Spam detection is a critical application of NLP that aims to automatically identify and filter unwanted, unsolicited, or harmful messages, especially in email and messaging platforms. Early spam detectors relied on rule-based filtering or statistical machine learning models like Naïve Bayes, which used handcrafted features such as keyword frequency, sender information, and metadata. As spam techniques evolved, modern systems adopted more robust machine learning and deep learning models capable of understanding contextual cues within text. Transformer-based encoders such as BERT analyze semantic patterns, suspicious phrasing, phishing indicators, and linguistic anomalies to classify messages with higher accuracy. The process typically includes preprocessing, vector representation of text, feature learning, and binary or multi-class classification. Spam detection is essential for protecting users from phishing attacks,

scams, malware links, and advertising overload. These systems improve communication efficiency, enhance cybersecurity, and reduce risks associated with fraudulent content. However, challenges persist in detecting sophisticated spam that mimics legitimate communication, handling multilingual spam, and adapting to continuously evolving adversarial tactics. Ongoing research in adversarial learning and contextual embedding aims to further strengthen spam detection systems.

5. Text Prediction – 7-mark Answer Text prediction refers to the automatic suggestion of the next word, phrase, or sentence based on previously typed or spoken text, and it is fundamental to modern NLP-driven writing and communication tools. Earlier systems relied on n-gram language models that estimated the probability of the next word using fixed-size context windows, but they struggled with long-range dependencies and contextual understanding. The advent of neural networks, particularly LSTMs and Transformer-based language models, revolutionized text prediction by enabling models to learn subtle semantic relationships, syntactic patterns, stylistic trends, and discourse-level coherence. During prediction, the model analyses the input context, generates probability distributions over the vocabulary, and selects the most suitable next token. Text prediction powers autocomplete in keyboards, coding assistants, search engine query suggestions, chatbots, and machine translation systems. It enhances typing speed, reduces errors, and improves user experience. Although powerful, text prediction must handle ambiguity, rare words, domain-specific vocabulary, and the risk of generating biased or incorrect suggestions. Continuous training on diverse datasets has steadily improved prediction accuracy and contextual awareness, making it a vital component of intelligent language applications.